

Interactive MObjects Tester: Algorithm and Parameters

Reference guide for the MObjects foreground-mask backend and tuning controls.

Scope: this document describes the new MObjects tester, not the SEP implementation. It keeps the same manifest, geometry, profile, and PNG review workflow so future comparisons are straightforward.

Algorithm Summary

MObjects detects astronomical sources by building a max tree of the preprocessed image and applying statistical attribute filtering to identify significant connected structures. The project README describes it as a tool for detecting sources, creating segmentation maps, and producing parameter tables; the implementation is based on the C work by Teeninga et al. and wrapped by Caroline Haigh's Python package.

1. Prepare either the residual image or original image as the detection surface.
2. Estimate or accept supplied background mean, variance, soft bias, and gain values.
3. Subtract the background, optionally smooth with a Gaussian kernel, truncate negative values, and replace NaNs for MObjects.
4. Build a max tree over the preprocessed image.
5. Run MObjects statistical filtering using `alpha`, `move_factor`, and `min_distance`.
6. Relabel the object map, restore non-finite pixels to background, and convert MObjects labels to a boolean foreground mask.
7. Apply comparison-layer filters: minimum area, maximum area, elongation, central exclusion, and mask dilation.

MObjects Parameters

Parameter	Default	Effect
<code>alpha</code>	1e-6	Significance level used by the MObjects statistical test. Smaller values are stricter.
<code>move_factor</code>	0.5	Controls how far object markers move upward in the tree.

		Higher values reduce spread of large objects.
min_distance	0.0	Minimum brightness separation required between objects.
gaussian_fwhm	2.0 px	FWHM used by MObjects preprocessing before tree construction; 0 disables smoothing.
soft_bias	0.0	Constant bias term used when estimating gain.
gain	-1.0	Electrons per ADU. A negative value asks MObjects to estimate it.
bg_mean	NaN	Background mean. NaN asks MObjects to estimate it from flat tiles.
bg_variance	-1.0	Background variance. A negative value asks MObjects to estimate it.

Post-Detection Parameters

Parameter	Default	Effect
minarea	5 px	Rejects small detected segments after MObjects relabeling.
max_area	230 px	Rejects broad segments that are more likely galaxy/bar structure than compact foreground objects.
max_elongation	6.0	Rejects very elongated detections using weighted second moments.
exclude_center_pixels	8 px	Rejects detections close to the galaxy centre, with arcsec display conversion available.
dilation_radius	2 px	Expands accepted segments before replacing/masking pixels.

Spike-Gate Parameters

The spike-gate run is retained as an optional comparison view. It runs MTOBJECTS with a usually more inclusive `move_factor`, then keeps only segments intersecting bar-major-axis spike samples.

Parameter	Default	Effect
<code>spike_gate_move_factor</code>	0.3	Move factor for the gate-run segmentation.
<code>spike_excess_fraction</code>	0.25	Required fractional excess above neighbouring bar-profile levels.
<code>spike_neighbour_inner_arcsec</code>	4.0	Inner radius of the local comparison annulus along the profile.
<code>spike_neighbour_outer_arcsec</code>	15.0	Outer radius of the local comparison annulus along the profile.
<code>spike_side_offset_samples</code>	3	Sample offset used to verify a local side drop.
<code>spike_side_drop_fraction</code>	0.4	Required excess over side samples.
<code>spike_window_samples</code>	2	Number of neighbouring profile samples added around detected spikes.

Operational Notes

- MTOBJECTS currently expects compiled shared libraries in the MTOBJECTS checkout; use `--mtobjects-root` or `MTOBJECTS_ROOT`.
- The tester treats MTOBJECTS as the detector and keeps comparison filters outside the library adapter.
- Mask replacement currently uses the median of finite unmasked pixels, matching the SEP tester's simple visualization-oriented cleaning.
- The MTOBJECTS README warns the project is a work in progress, so parameter sweeps should record library version or commit when used for comparisons.

Sources

- [CarolineHaigh/mtobjects README](#): source detection, segmentation maps, dependencies, and command-line parameters.
- [CarolineHaigh/mtobjects source files](#): `main.py`, `preprocessing.py`, `tree_filtering.py`, `maxtree.py`, and `postprocessing.py`.
- Local implementation: `interactive_mtobjects_parameter_tester.py`.